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Variations in The Risk of Cerebral Ischemic Complications Post Blunt Cerebrovascular Injuries

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Introduction: Early treatment of blunt cerebrovascular injuries (BCVI) with anticoagulation or antiplatelet is strongly recommended to prevent cerebral ischemic accidents (CVA). We aimed to assess risk factors of CVA in patients with BCVI.

Methods: Using the Healthcare Cost and Utilization Project's 2018 Nationwide Readmission Database, we identified patients ≥18 years of age with BCVI who were admitted between January and October. Our primary outcome was CVA at any time post-injury. We performed a Lasso regression analysis with adaptive selection to generate a predictive model with least deviance using patients' demographics, injury severity and pattern, and anatomic variations of BCVI, such as involved vessels and bilaterality. Weighted multiple logistic regression was performed to obtain population estimates of adjusted odds ratios (OR) of each element in the model.

Results: Our cohort included 1,470 patients; of whom 128 patients had CVA during index admission (8.7%). Although readmission rate was high (n=265, 19.9%), only 8 patients (0.6%) were readmitted with CVA diagnoses. Majority of BCVI involved the vertebral arteries (n=1,105, 77.2%) then carotid arteries (282, 19.7%), then combined (n=44, 3.1%). Combined injury had highest risk of CVA (13.6% versus 11.4% for carotid injury and 8.1% for vertebral injury). This was not statistically significant. Using Lasso regression modeling, the model obtained using adaptive selection had the least deviance and therefore was selected. Carotid artery injury had highest adjusted odds of CVA (OR: 3.22, 95%CI: 1.39-7.47), while vertebral artery injury had an OR of 2.42 (95%CI: 1.03-5.67). High grade injuries had significantly higher odd of CVA (OR: 2.92, 95%CI: 1.35-6.33). Severe injury (ISS≥ 16) was associated with lower odds of CVA (OR: 0.49, 95%CI: 0.33-0.73). Using weighted analysis, only carotid injury and age had significantly higher odds of CVA (OR: 3.29, 95%CI: 1.07 – 10.14, and OR: 1.01, 95%CI: 1.00-1.03, respectively).

Conclusion: BCVI is associated with high risk of stroke, although this risk is variable depending on anatomic location and severity. Delayed cerebrovascular ischemic complications are very rare. Future validation of these findings using more granular date could allow for an optimal risk stratification and tailored management.